

Tut Test 6

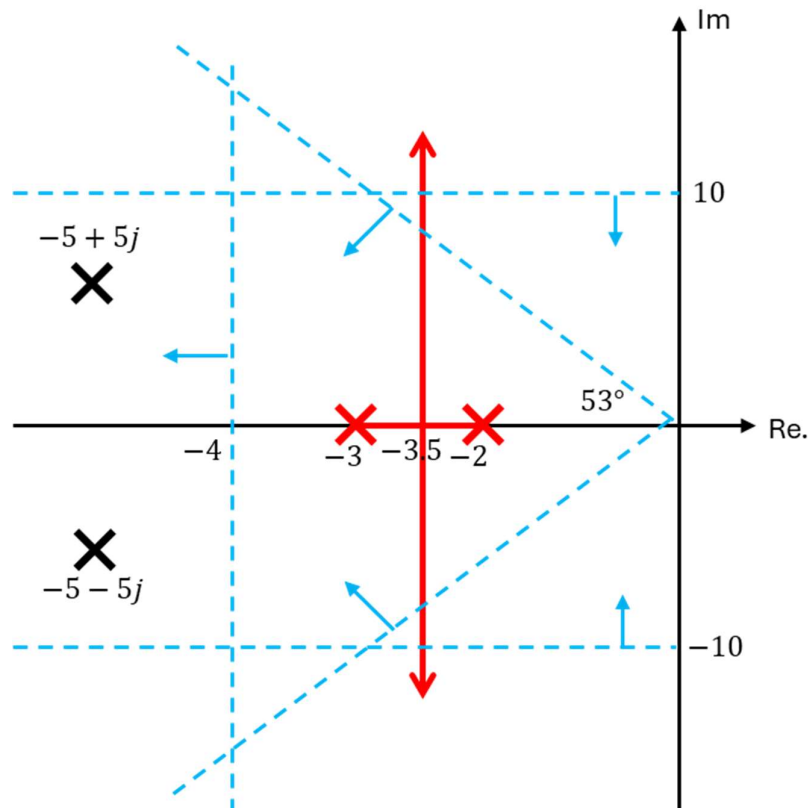
Consider the system with the transfer function $G(s) = \frac{18}{(s+2)(s+3)}$

This system is to be placed in negative feedback with a lead compensator to satisfy the constraints:

- i. Settling time decreased to 50% compared to the open-loop system.
 - ii. Damping ratio of at least 0.6.
 - iii. Damped frequency of oscillation of less than 10 rad/s
 - iv. Accuracy with respect to position setpoints of more than 90%
- a) Draw the root locus of the system, and indicate the boundaries within which the transient response specifications i-iii will be satisfied, and hence, confirm that a complex pole pair at $-5 \pm 5j$ meets these requirements. [8 marks]
 - b) Why is it not possible to satisfy the specifications with proportional control? [2 marks]
 - c) Design a lead compensator for the system that places the dominant closed-loop poles at the suggested location $(-5 \pm 5j)$. State whether your design also satisfies specification iv, and if not, give a brief suggestion of how you can change your design to achieve greater accuracy. [15 marks]

Solutions:

- a) The root locus of the system is shown below. [2 marks]



i) The open-loop system is overdamped so its settling time will be determined by the time constant of the slowest pole. The slower pole is at -2, giving $\tau = 0.5$ seconds, so the dominant pole or poles in closed loop should have $\tau \leq 0.25$, meaning that they must be further to the left than -4. [2 marks]

ii) Damping ratio of 0.6 corresponds to poles on line of angles $\pm \arccos(0.6) = \pm 0.92 \text{ rad} = \pm 53^\circ$ with respect to the negative real axis. [1 mark]

iii) Damped frequency of oscillation of 10 rad/s can be represented by horizontal lines intersecting the imaginary axis at $j\omega = \pm 10 \text{ rad/s}$. [1 mark]

The suggested pole pair is within all specification boundaries so it meets the requirements. [1 mark]

- b) There are no points on the uncontrolled system's root locus that satisfy the speed specification, so it is not possible to obtain a closed-loop pole meeting all requirements with proportional control.
- c) For the desired pole $s_d = -5 + 5j$ to be available on the root locus, the angle criterion must be satisfied at that point i.e. $\angle C(s_d)G(s_d) = \pm 180^\circ + \dots$

The angles to s_d from the open-loop poles are:

$$\pi - \arctan\left(\frac{5}{3}\right) = 2.11 \text{ rad} = 121^\circ \text{ for the pole at -2 [1 mark]}$$

$$\pi - \arctan\left(\frac{5}{2}\right) = 1.95 \text{ rad} = 112^\circ \text{ for the pole at -3 [1 mark]}$$

This gives a total angular contribution of -233° for the system poles, so the compensator must add 53° to get to the nearest odd multiple of 180 at -180° . [1 mark]

The zero will add this angle by itself when it is at $-5 - \frac{5}{\tan(53^\circ)} = -6.3$, so the zero should be somewhere between -6.3 and -5. [2 marks]

Any zero in this range is acceptable, but for this sample solution we will try a zero at -6. [1 mark]

This gives a zero angle of $\arctan\left(\frac{5}{1}\right) = 78.7^\circ$ [1 mark]

So the pole must subtract $78.7 - 53^\circ = 25.7^\circ$ [1 mark]

This puts the pole at $-5 - \frac{5}{\tan(25.7)} = -15.29$ [1 mark]

The compensator is then $C(s) = \frac{0.1667s + 1}{0.06542s + 1}$ [1 mark]

From the magnitude criterion, the gain required to have a closed-loop pole at the desired location is then $K = \frac{1}{|C(s_d)G(s_d)|} = 1.53$ [1 mark]

Using the final value theorem on the error, the final value required for less than 10% error with respect to constant inputs is $\frac{1}{1+KG(0)} \leq 0.1 \therefore KG(0) \geq 9$. The gain of the system is 3, so the gain of the controller must be at least 6. [2 marks]

This is more than the gain obtained, so the system will not satisfy the accuracy requirement. A larger gain could be obtained by selecting a zero that is further to the left, so the pole and zero of the compensator are further apart, or by adding a lag compensator in cascade. [2 marks]